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Project Report

Submitted To

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Course Code	530220
Project Name	Online Shop Management System (OSMS)

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Abstract

The **Online Shop Management System (OSMS)** is a web-based e-commerce platform designed to simplify online shopping for customers and streamline backend operations for administrators. The system allows users to browse, search, and purchase products from different categories such as electronics, apparel, and furniture. It ensures a smooth shopping experience with secure payment integration, real-time order tracking, and an intuitive user interface.

For administrators, OSMS provides centralized control over product management, order handling, customer data, and analytics through a dedicated dashboard. The system has been developed using **Python (Django Framework)** for the backend and **HTML, CSS, Bootstrap and JavaScript** for the frontend, with **SQLite** as the database.

The project follows a structured software development approach—beginning with requirement analysis, system design (including DFD, E-R, and UML diagrams), and continuing through implementation, testing, and deployment. Key modules include **User Management, Product Management, Cart, Order, and Payment.** Various testing methods such as unit, integration, and user acceptance testing were conducted to ensure system reliability and performance.

The OSMS provides a scalable and secure foundation for online retail operations, suitable for small and medium-scale businesses. Future enhancements may include developing a mobile app, integrating AI-based product recommendations, multi-language support, real-time delivery tracking, and cloud-based optimization.

Overall, this project demonstrates a complete end-to-end solution for managing online shopping activities efficiently, combining usability, performance, and modern web technologies to deliver a robust e-commerce experience.

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ONLINE SHOP MANAGEMENT SYSTEM

1. Introduction

In today's digital era, most business transactions and purchases are shifting from traditional physical markets to online platforms. Managing these online operations efficiently requires an integrated system that can handle customers, inventory, sales, and payments in real time. The **Online Shop Management System (OSMS)** has been developed to meet these modern commercial needs by providing a single platform for both customers and administrators. Customers can easily search for products, compare prices, and place orders online, while administrators can manage the entire store from product listings and order tracking to payment and delivery management.

Overall, the Online Shop Management System represents a step toward the digital transformation of traditional shopping methods, offering convenience, efficiency, and transparency to both buyers and sellers.

1.1 Problem Definition

Shopping physically for multiple items can be time consuming and inconvenient. The **Online Shop Management System** offers a centralized solution that enables customers to access diverse products, make purchases, and track orders easily.

1.2 Purpose

The main goal of the **Online Shop Management System** is to build a robust and user-friendly e-commerce platform that simplifies online shopping for customers and backend management for administrators. Specific **objectives** include:

- **Efficient Product Management:** Admins can easily add, update, or delete items such as electronics, apparel, or furniture.
- **Seamless Shopping Experience:** Customers can quickly search, filter, and browse items.
- **Secure Online Payments:** The system supports multiple online payment options, ensuring transaction safety.
- **Order and Inventory Tracking:** Admins have access to real-time data on stock levels and customer orders.
- **Returns and Exchanges:** A structured policy allows for hassle-free returns and exchanges within a designated time frame.
- **Centralized Control:** All store operations are managed through a unified dashboard, ensuring easy oversight and coordination.

1.3 Scope

The scope of this system includes everything necessary for operating a professional and functional online store. This covers:

- **Product Management:** Administrators can maintain detailed catalogs with product descriptions, prices, and multimedia content.
- **Customer Management:** Users can register, log in, manage their profiles, and access loyalty or promotional programs.
- **Order Handling:** Customers can add products to a shopping cart, place orders, and receive confirmations. Admins can track and fulfill these orders efficiently.
- **Payment Integration:** The system connects with secure payment gateways to facilitate smooth and reliable transactions.
- **Analytics and Reports:** Insights into user behavior, sales trends, and performance metrics help guide business decisions.
- **Additional Features:** Depending on system complexity, it may also support features like customer reviews, product recommendations, and integration with external software like accounting tools.

1.4 User and Literature Survey

To design an effective Online Shop Management System, it is essential to understand existing solutions and user expectations. A literature survey includes an analysis of similar platforms (e.g., Shopify, WooCommerce, Magento) to identify common features, strengths, and weaknesses. Simultaneously, user surveys and feedback collection help in understanding desired functionalities, pain points in current systems, and suggestions for improvement. This combination of research ensures the development of a system tailored to both market standards and user needs.

2. Analysis and Design

This phase explains how the system works and how different parts are connected.
It includes studying what the system needs (Requirement Analysis) and how it should be structured (Requirement Design).

2.1 Requirement Analysis

Requirement analysis helps identify the exact needs of both the administrator and the customer. It defines what features the system should include and what performance or quality standards it should maintain.
In this phase, we explore two key aspects of requirements. **Functional** and **Non-Functional** requirements, which together form the foundation of the project design.

2.1.1 Functional Requirements

Functional requirements describe what the system must do to perform its intended tasks. In **OSMS**, these include all the operations that allow users to interact with the system, such as registration, browsing, ordering and payment.

Below is a summary of the key functional requirements:

Sl. No.	Requirement	Description
1	**User Registration/Login**	Customers can register, log in, and manage their profiles securely.
2	**Product Management**	Admin can add, update, or remove products and categories.
3	**Shopping Cart**	Users can add products to a cart, view totals, and update quantities.
4	**Order Management**	The system manages order placement, tracking, and cancellation.
5	**Payment Integration**	Supports secure and seamless online payment processing.
6	**Reports and Analytics**	Admin can view sales data, user statistics, and system performance.

2.1.2 Non-Functional Requirements

Non-functional requirements define how the system should perform rather than what it should do.
They ensure the platform is efficient, secure, reliable, and user-friendly — qualities essential for an online shopping platform like OSMS.

Sl. No.	Requirement	Description
1	**Usability**	The interface should be attractive, simple, and easy to use for all users.
2	**Performance**	The website must respond quickly and handle multiple requests efficiently.
3	**Security**	Sensitive data like passwords and payments must be securely encrypted.
4	**Reliability**	The system should perform consistently without data loss or crashes.
5	**Scalability**	The platform should support future growth in users and products.
6	**Maintainability**	The system should be easy to maintain, debug, and upgrade.

2.2 Requirement Design

Once the requirements are analyzed, the next stage is to design how the system will actually work.

The design focuses on visualizing the flow of data, user interaction, and internal processes through diagrams such as the Context Diagram, Data Flow Diagram (DFD), Entity Relationship (ER) Diagram, and various UML diagrams (Sequence, Activity, Class, and Use Case).

These diagrams together define the logical and functional structure of the Online Shop Management System.

2.2.1 Context Diagram of OSMS

The context diagram represents the system as a single process and shows how it interacts with external entities such as users, administrators, and payment gateways. It helps identify the system's boundaries and data exchanges.

In OSMS, the external entities are:

- **Customer** → browses products, places orders, makes payments
- **Admin** → manages inventory, orders, and user accounts
- **Payment Gateway** → processes and verifies online payments

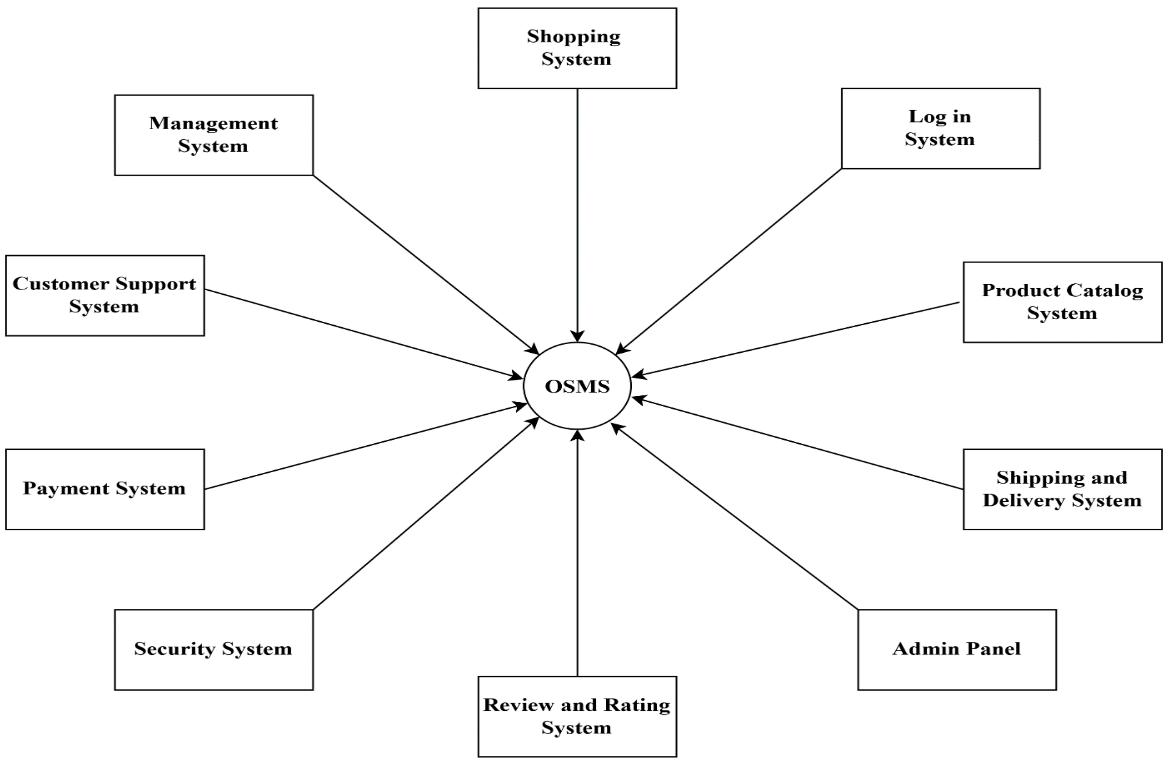


Fig: 2.2.1 Context Diagram of OSMS

2.2.2 Data Flow Diagram (DFD)

The Data Flow Diagram shows how data moves through the system. It provides a detailed look at inputs, processes, and outputs.

For OSMS:

- **Level 0 (Context Level):** Overall data exchange between the user, admin, and the system.
- **Level 1:** Explains core operations such as "View Product," "Add to Cart," "Make Payment."
- **Level 2:** Breaks down each process into internal operations like database updates and verification

➔ **DFD Level-0:**

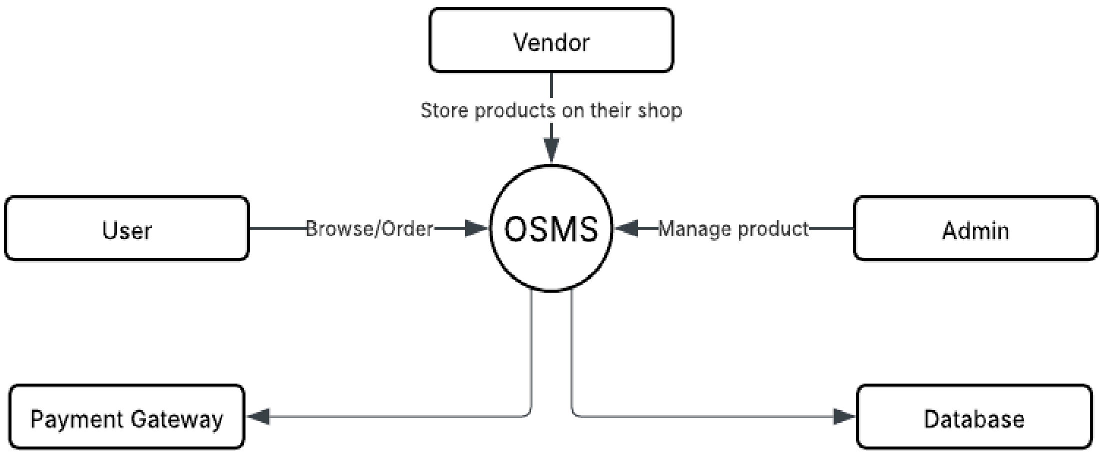


Fig: 2.2.2 Data Flow Diagram (DFD Level-0)

➔ **DFD Level-1**

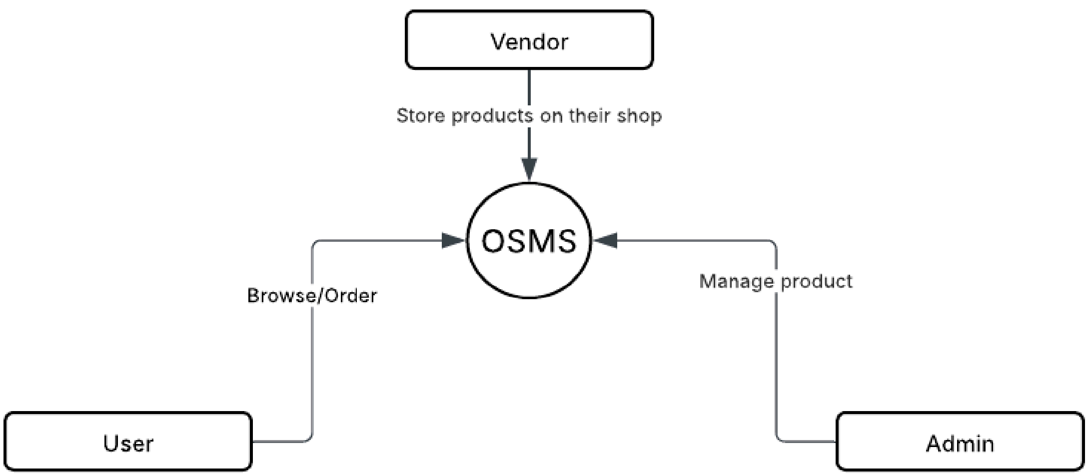


Fig: 2.2.2 Data Flow Diagram (DFD Level-1)

➔ DFD Level-2

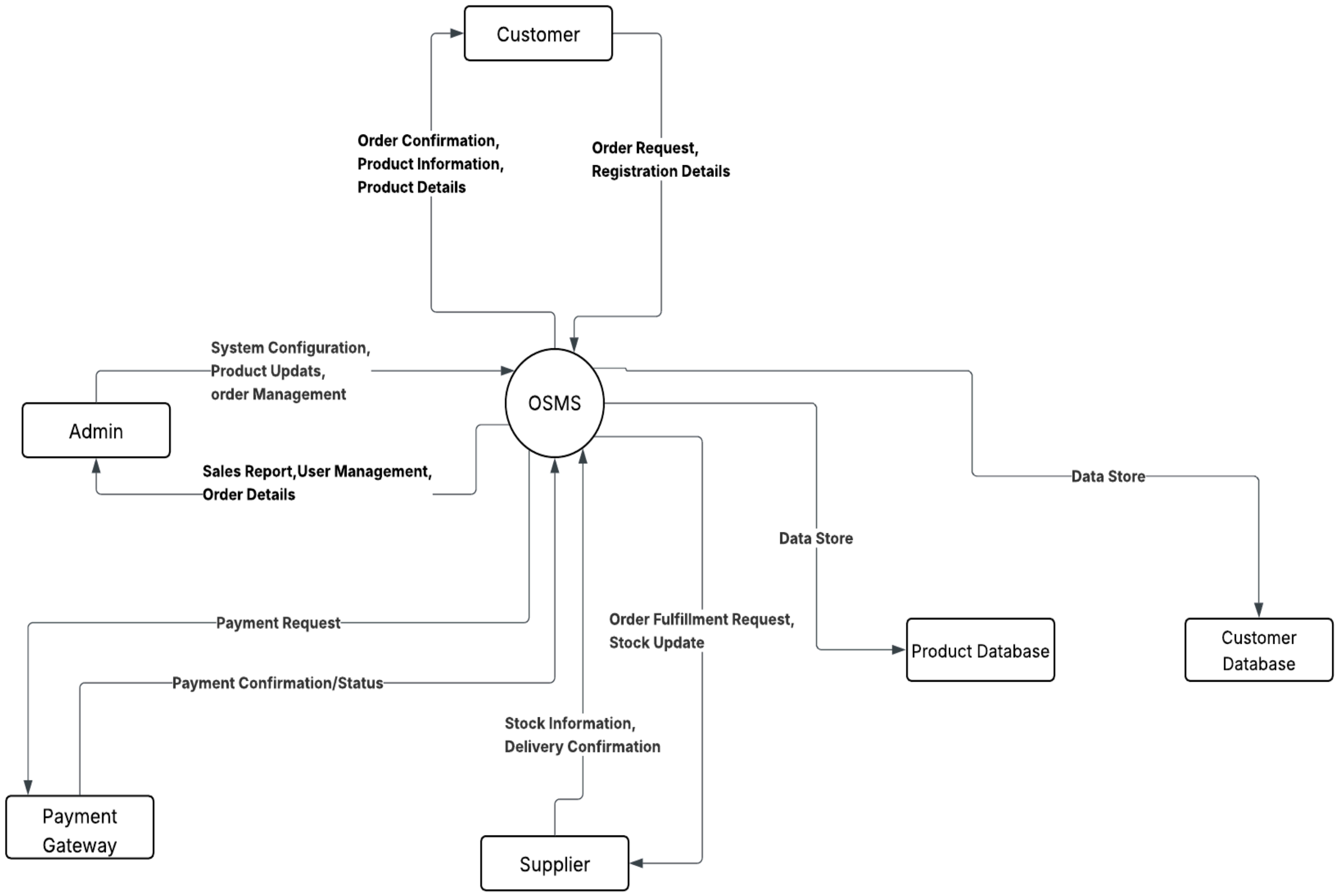


Fig: 2.2.2 Data Flow Diagram (DFD Level-2)

2.2.3 E-R Diagram

The E-R Diagram models how data entities relate to each other in the system's database.
In OSMS, key entities include:

- **User** (stores user information)
- **Product** (details about each item)
- **Order** (links users with purchased products)
- **Payment** (stores transaction details)

The relationships among them ensure data consistency and efficient retrieval.

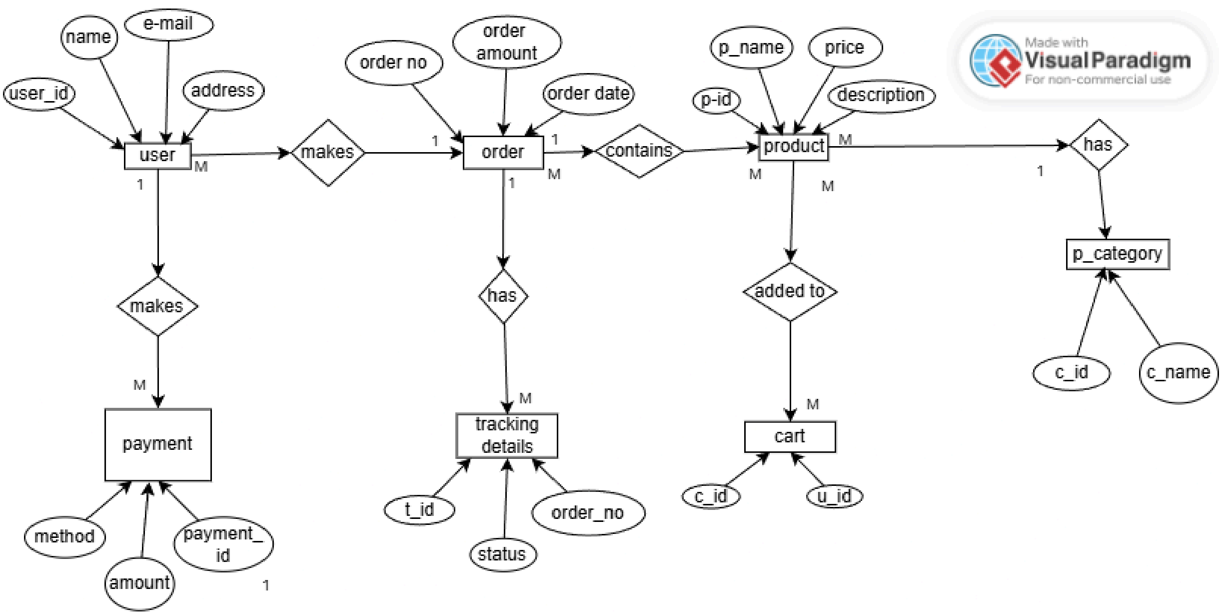


Fig: 2.2.3 E-R Diagram

2.2.4 Sequence Diagram

The sequence diagram illustrates how different parts of the system communicate step-by-step.
For example, when a customer places an order:

1. The system retrieves product details.
2. The user confirms the order.
3. Payment is processed.
4. The order is saved and confirmed.

This diagram visually represents the message flow between the user, the system, and the database.

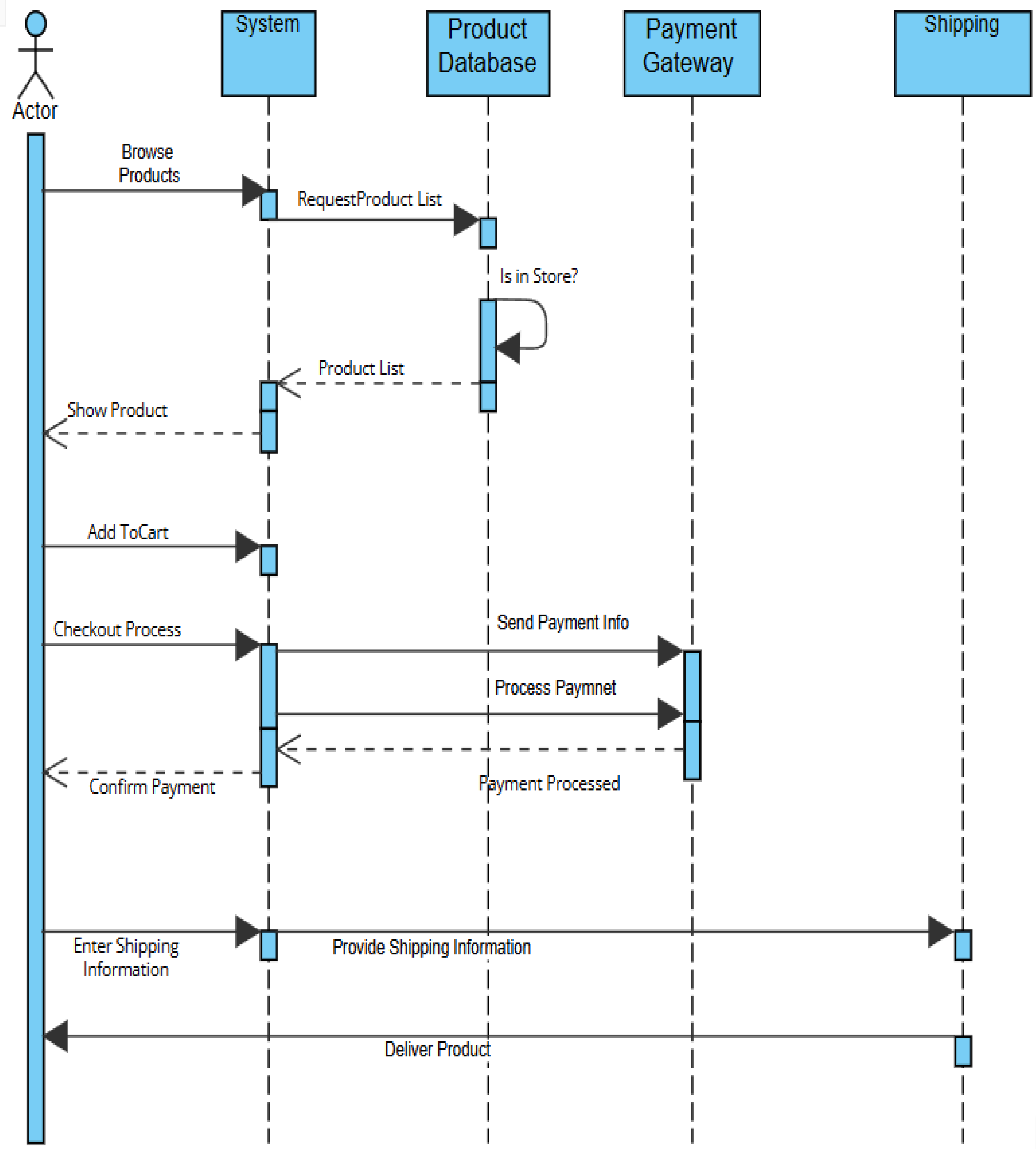


Fig: 2.2.4 Sequence Diagram

2.2.5 Activity Diagram

The activity diagram shows the workflow of actions or activities in a process. In the OSMS, it represents how a user moves through the system from login, product browsing, adding items to cart, confirming orders, making payments and finally logging out.

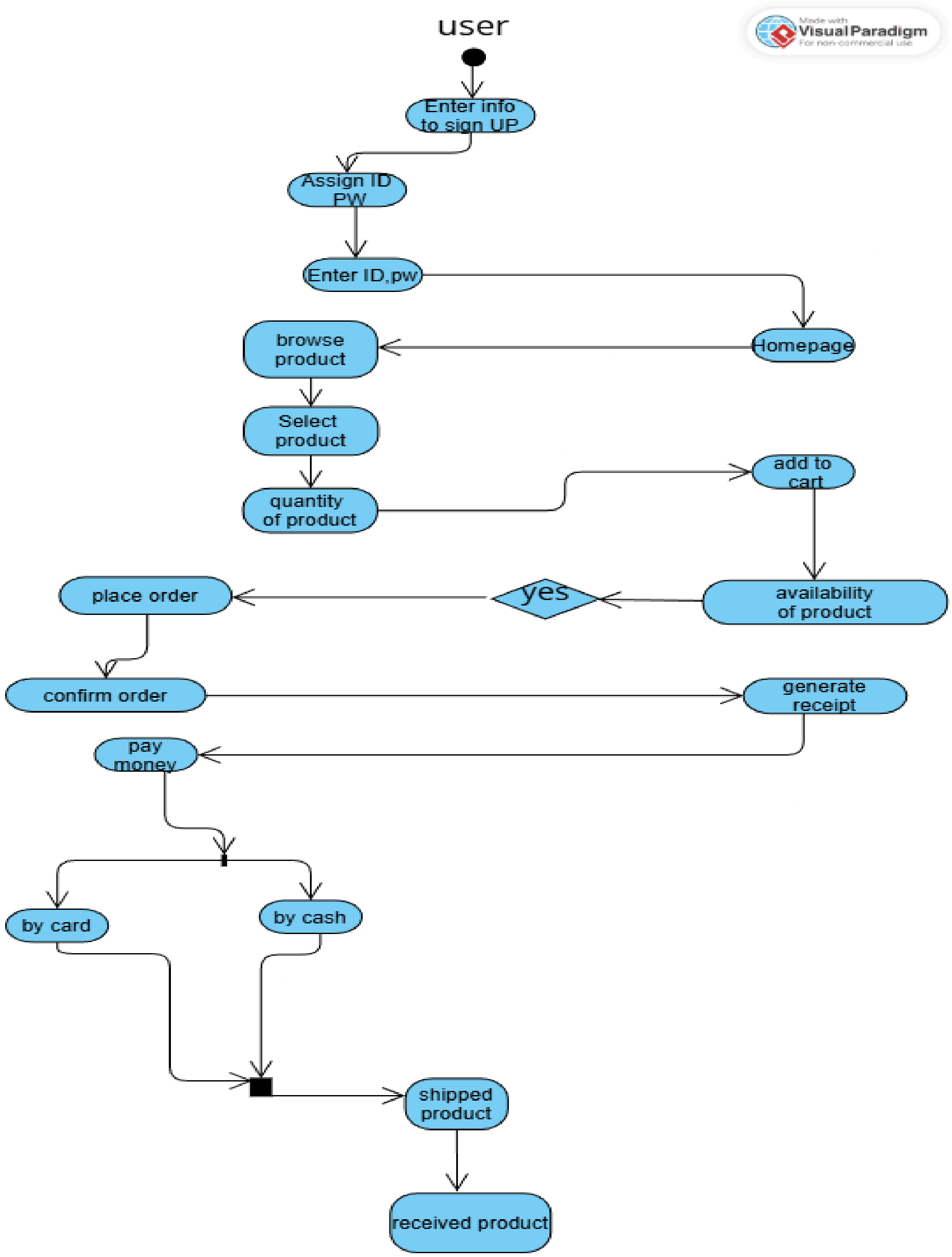


Fig: 2.2.5 Activity Diagram

2.2.6 Class Diagram

The class diagram displays the static structure of the system. It shows the main classes, their attributes, methods and relationships. In OSMS, the main classes are **User**, **Product**, **Cart**, **Order** and **Payment**. This helps developers understand how objects are related and interact programmatically.

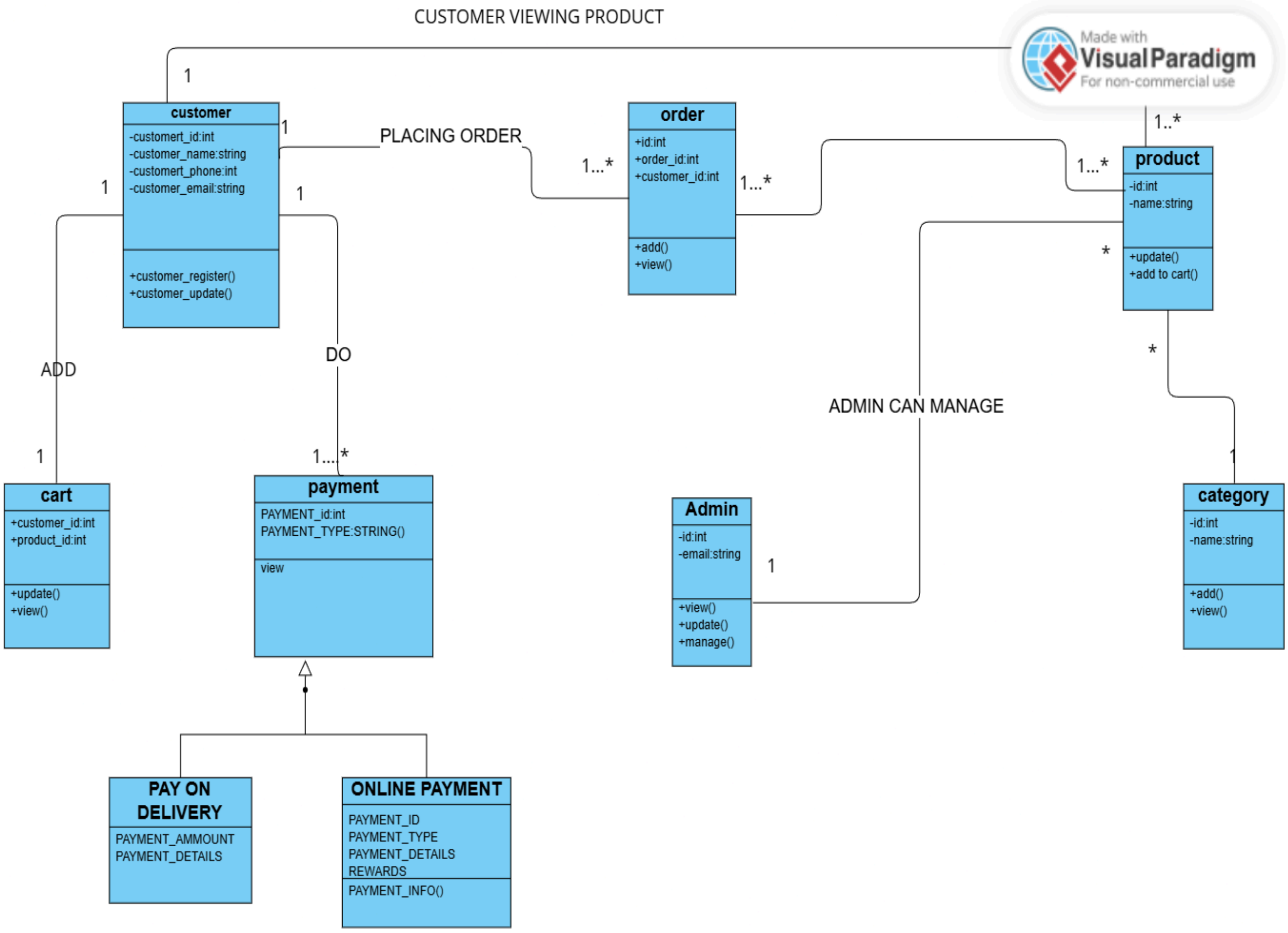


Fig: 2.2.6 Class Diagram

2.2.7 Use Case Diagram

The use case diagram shows the overall interactions between actors (users) and the system. For OSMS:

- **Actors:** Customer and Admin
- **Customer Use Cases:** Register, Login, Search Product, Add to Cart, Make Payment, Track Order
- **Admin Use Cases:** Add Product, Manage Orders, View Reports, Manage Users

It provides a complete picture of what actions are possible and who performs them.

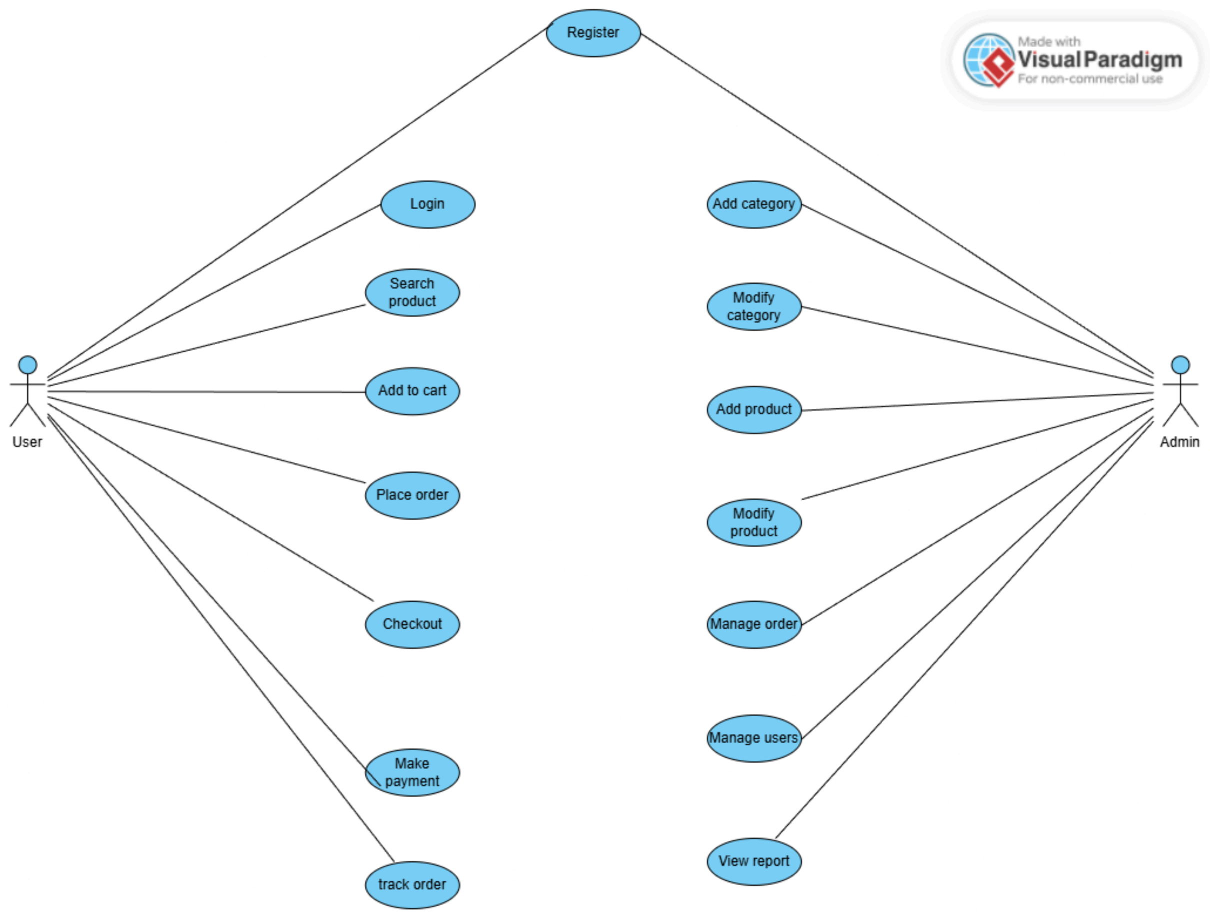


Fig: 2.2.7 Use Case Diagram

3. System Requirements

This section lists the minimum hardware and software required to develop and run the **Online Shop Management System (OSMS)** smoothly.

Proper configuration ensures good performance and stability of the system.

3.1 Hardware Requirements

Component	Minimum Specification
Processor	Intel Core i3 or higher
RAM	4 GB or more
Hard Disk	100 GB free space
Monitor	15” color display
Keyboard & Mouse	Standard input devices
Internet Connection	Required for online operations

3.2 Software Requirements

Component	Specification
Operating System	Windows 10 / 11 or Linux (Ubuntu)
Programming Language	Python (Django Framework)
Database	SQLite
Frontend Technologies	HTML, CSS, Bootstrap, JavaScript
IDE / Editor	VS Code or PyCharm
Browser	Google Chrome or Mozilla Firefox
Web Server	Django built-in server / Apache

4. Implementation

This phase focuses on building and executing the actual system based on the design and requirements.
All modules of the **Online Shop Management System (OSMS)** are developed using **Python (Django)** for the backend and **HTML, CSS, Bootstrap** for the frontend. The database is managed using **SQLite**

4.1 Module Description

Module Name	Description
Admin Module	Allows the admin to manage products, view orders, and handle user information.
User Module	Enables customers to register, log in, and manage their profiles.
Product Module	Displays all available products with details such as name, price, and category.
Cart Module	Lets users add, update, or remove items before checkout.
Order Module	Handles order placement, payment, and tracking.
Payment Module	Manages secure online transactions and payment verification.

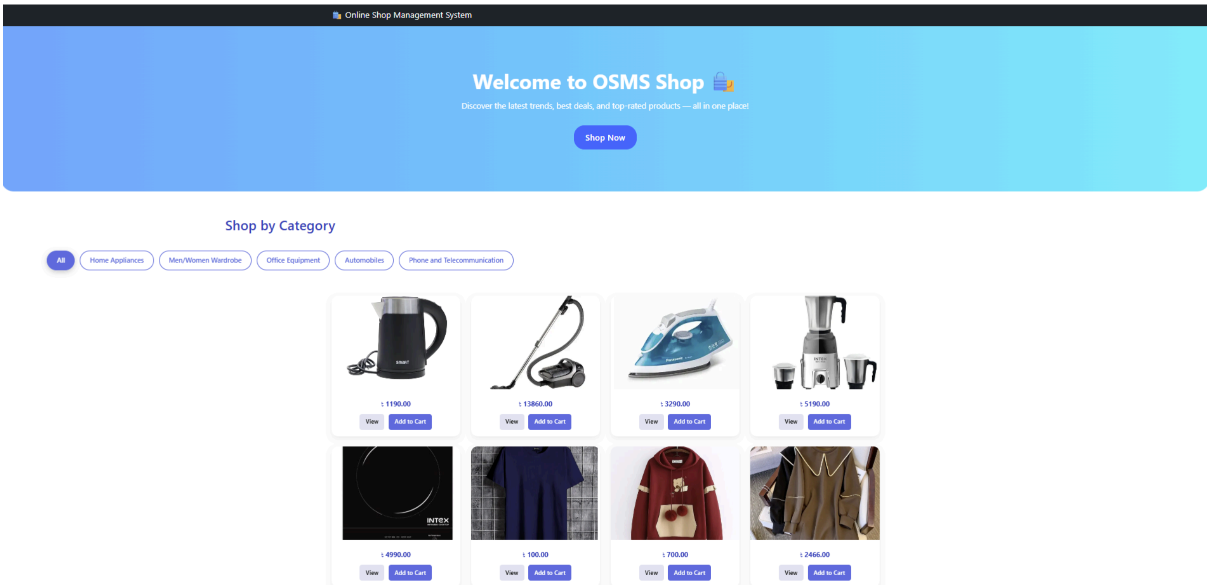
4.2 Implementation Tools

- **Backend:** Python (Django Framework)
- **Frontend:** HTML, CSS, Bootstrap
- **Database:** SQLite
- **Editor:** VS Code
- **Server:** Django built-in server

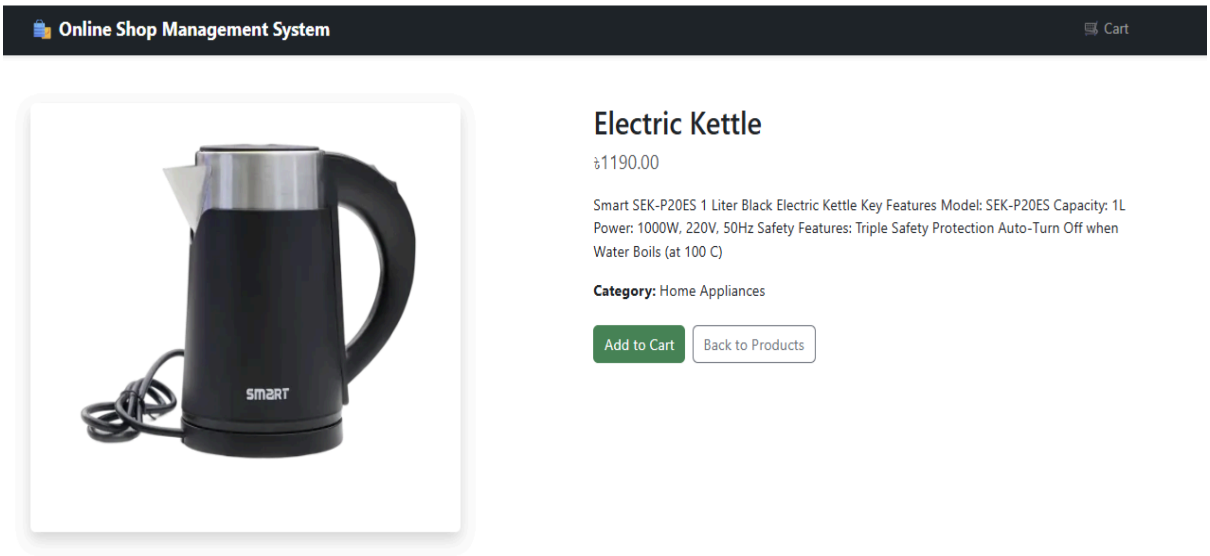
4.3 Screenshots

Screenshots of the implemented system can include:

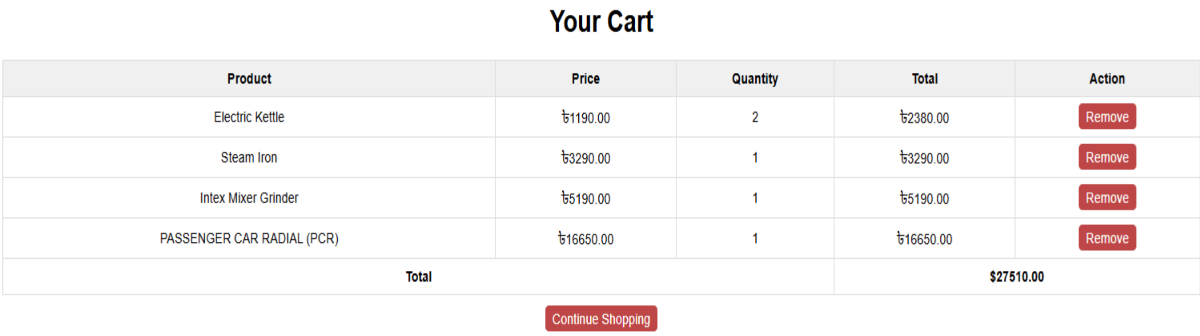
1. **Homepage / Product Page**



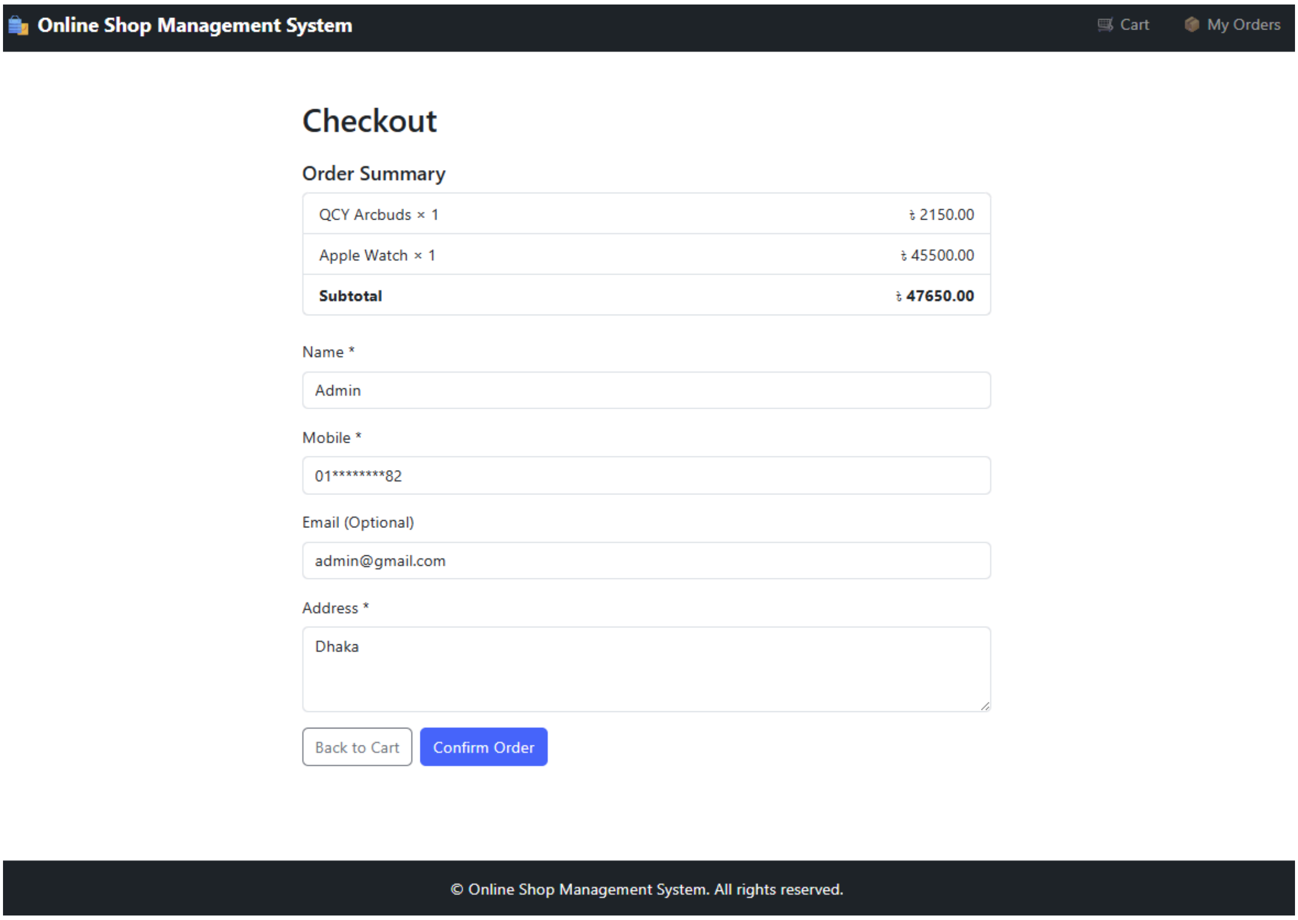
2. **Product Details Page**



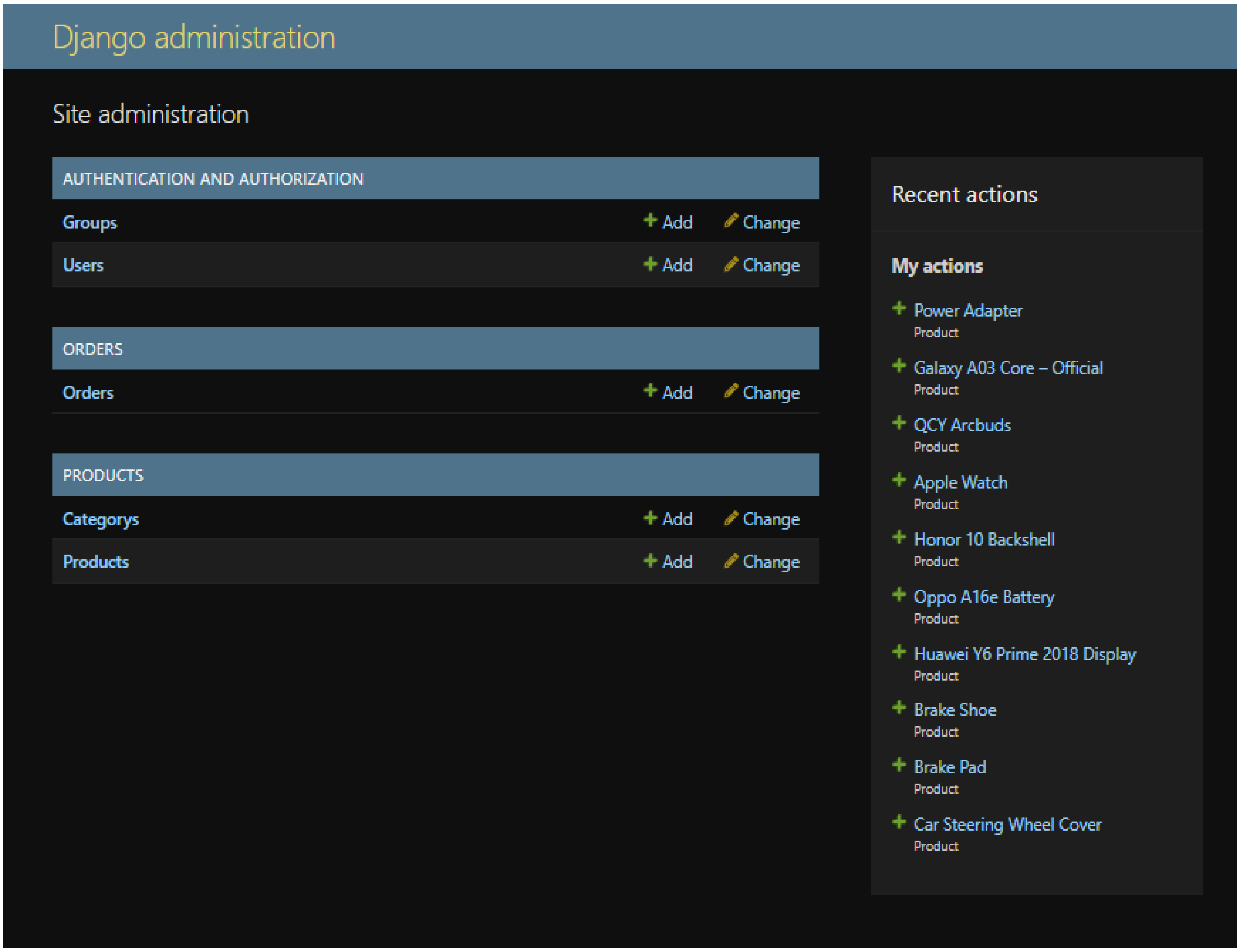
3. Shopping Cart



4. Checkout Page



5. Admin Dashboard



5. Testing

Testing is performed to ensure that all parts of the **Online Shop Management System (OSMS)** work correctly and meet the specified requirements.
It helps identify and fix errors, verify system performance, and confirm that all modules function together properly.

5.1 Types of Testing Used

Testing Type	Description
Unit Testing	Each module (e.g., login, cart, payment) is tested individually to verify correct operation.
Integration Testing	Checks whether all modules work together as expected after being combined.
System Testing	Tests the complete system for overall functionality and performance.
User Acceptance Testing (UAT)	Conducted by users to ensure the system meets real-world needs and expectations.

5.2 Testing Results

During testing, all modules were executed successfully without major errors.
Functions like **user login**, **product search**, **cart update** and **payment confirmation** worked as expected. Minor design adjustments were made for better responsiveness and user experience.

5.3 Screenshots

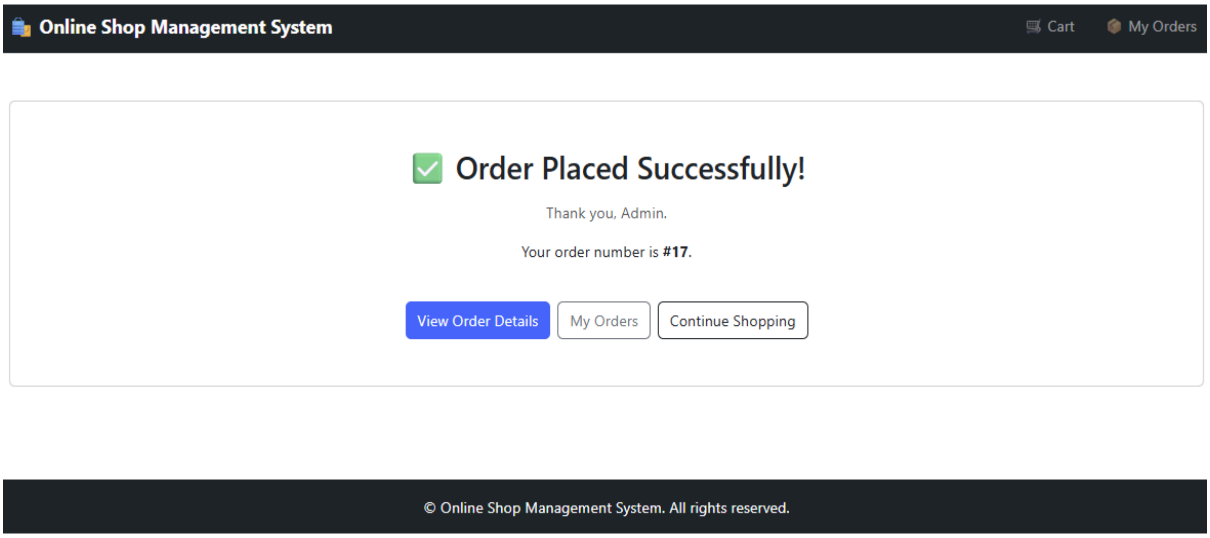
- **Product added to cart**

Your Cart

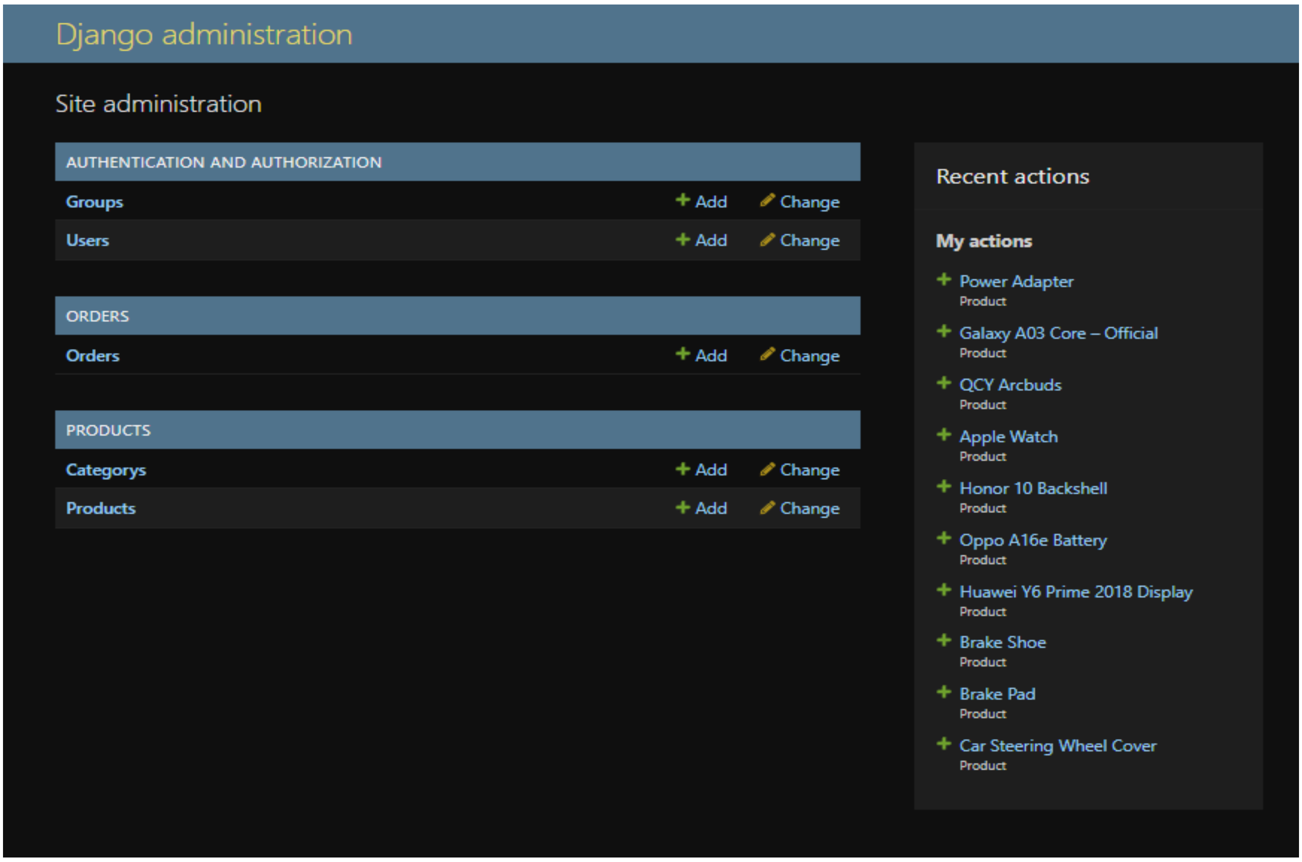
Product	Price	Quantity	Total	Action
Electric Kettle	৳1190.00	2	৳2380.00	Remove
Steam Iron	৳3290.00	1	৳3290.00	Remove
Intex Mixer Grinder	৳5190.00	1	৳5190.00	Remove
PASSENGER CAR RADIAL (PCR)	৳16650.00	1	৳16650.00	Remove
Total			$27510.00	

Continue Shopping

- **Order Placement Confirmation**



- **Admin dashboard view**



6. Deployment

Deployment is the process of making the project live and accessible for real users. After testing, the **Online Shop Management System** can be deployed on a **local server** for demonstration or a **cloud server** for public access.

6.1 Deployment Steps

1. Install required dependencies (Python, Django, database, etc.).
2. Apply migrations and load initial data.
3. Run the server locally using the Django command `python manage.py runserver`.
4. For live hosting, upload the project to platforms like **PythonAnywhere**, **Heroku**, or **AWS**.
5. Test the deployed system to ensure all pages and functions work properly.

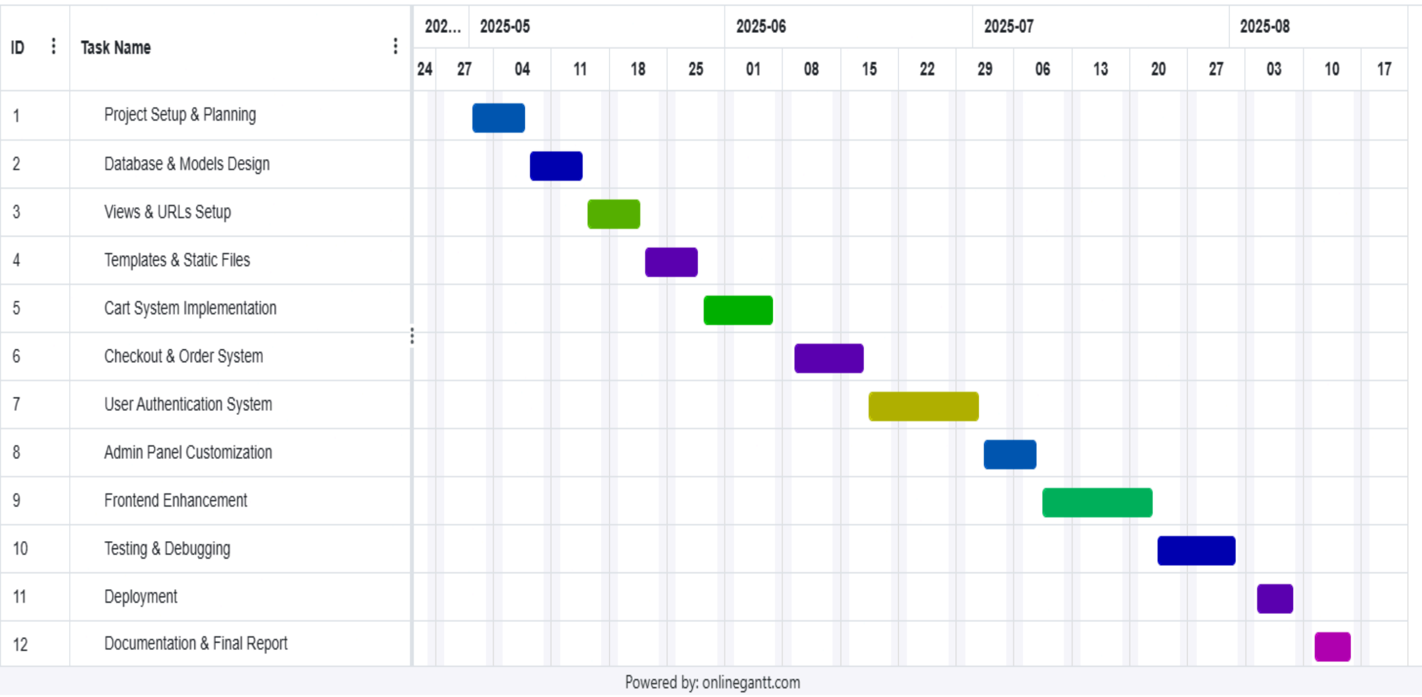
6.2 Accessing the System

- **Admin Interface:** Used for product and user management.
- **User Interface:** Accessible through a browser for product browsing, ordering, and payment.

The system can be accessed from any device with an internet connection.

6.3 Project Timeline

The development of the **Online Shop Management System (OSMS)** followed a structured and time-bound schedule managed through a Gantt chart. This visual tool assisted in tracking task progress, identifying dependencies and ensuring that all activities were completed within their designated deadlines.



Key Milestones

Based on the defined schedule, several key milestones were identified throughout the project lifecycle:

- **Milestone 1: Project Kick-off and Planning (24th April 2025):** The project officially began with scope definition, task scheduling, and resource allocation. The technology stack and development tools were finalized during this stage.
- **Milestone 2: System Design & Setup (08th May 2025):** The database models, wireframes, and initial layout templates were designed. The foundational structure for the backend and frontend setup was completed.
- **Milestone 3: Core Development Phase (02th June 2025):** This phase focused on developing the essential backend modules such as user authentication, product management, and order system implementation.
- **Milestone 4: Integration & Customization (27th June 2025):** The frontend components were integrated with backend APIs. Admin panel customization and user interface refinement were completed for a seamless user experience.
- **Milestone 5: Testing & Deployment (18th July 2025):** Rigorous system testing was conducted to identify and fix bugs. After successful validation, the project was deployed to a live server environment for demonstration.
- **Milestone 6: Documentation & Final Submission (05th August 2025):** The project report, technical documentation, and user manual were finalized and submitted for evaluation.

7. Future Scope

The current system performs all essential operations of an online store.
However, it can be improved in the future to enhance user experience and system performance.

Possible Future Enhancements

- Develop a **mobile app** version for Android and iOS users.
- Add **AI-based product recommendation** features.
- Implement **multi-language support** for global users.
- Integrate with **external courier and delivery APIs** for live order tracking.
- Add **review and rating systems** for better user engagement.
- Use **cloud database and caching** for faster performance.

8. References

- Django Documentation - https://docs.djangoproject.com/
- W3Schools - https://www.w3schools.com/
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- Stack Overflow - for coding and debugging support
- YouTube - Django E-commerce Project Tutorials